

Diego Barrio Hortigüela

Senior Backend Engineer · Java, Spring & Applied AI

Payments and banking platforms · Event-driven systems · Applied AI

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About

Senior Backend Engineer with over 15 years in Java, most of them in banking and payments. I currently work on the payment and billing APIs of Europcar Mobility Group, which handle around 10,000 transactions and millions of euros a day, on Java 21, Spring Boot, Kafka and Kubernetes. Before that I spent eight years on Banco Sabadell systems: mortgage simulation and amortization in core banking, ATM services, and later the security of the document platform behind its online banking. I work with DDD, hexagonal architecture and TDD. In my own projects I build LLM features where the model's output is validated against a domain contract before the system accepts it.

Core Competencies

Backend Engineering

Java 21 · Spring Boot · Spring Security · JPA / Hibernate · JDBC · REST APIs · Batch Processing · TDD

Software Architecture

Microservices · DDD · Hexagonal Architecture · Clean Architecture · Event-Driven · API Design

Domains

Payments · Billing & Invoicing · Core Banking · Mortgage Simulation · ATM Channel · Online Banking

Security

Spring Security · SAML · JWT · OAuth 2.0

Platform & Observability

Docker · Kubernetes · GitLab CI/CD · Jenkins · AWS · Azure · Splunk · Grafana · Prometheus · OpenTelemetry

Data & Messaging

Oracle · PostgreSQL · SQL Server · Flyway · Apache Kafka

AI Engineering

LLM Integration · RAG · Tool Calling · MCP · Agentic Workflows · Ollama · Claude Code

Professional Experience

Senior Backend Engineer

Mindden, embedded in Europcar Mobility Group's payments team · Mar 2024 – Present

- Build and maintain the payment and billing APIs, with ~10,000 payment transactions and millions of euros processed per day, averaging ~1,000 transactions an hour with higher peaks in high season.
- Daily billing runs issue tens of thousands of invoices with sub-second generation time, monitored through Splunk dashboards.
- Event-driven integration with Apache Kafka for asynchronous processing between services.

- OAuth 2.0 authentication for the payment API's clients, with Spring Cloud OpenFeign clients wired through Spring Boot auto-configuration.
- Services on Java 21, with the first microservices already migrated to Java 25.
- Drive DDD, hexagonal architecture and TDD in the team; GitLab CI/CD to Kubernetes on AWS.

Technologies: Java 21, Java 25, Spring Boot, OpenFeign, OAuth 2.0, Kafka, Kubernetes, AWS, Oracle, Splunk, GitLab CI/CD

Senior Analyst

Accenture, for Banco Sabadell · Jul 2022 – Mar 2024

- Responsible for the Document Management System (DMS) of Banco Sabadell's online banking, the platform that shares sensitive documents with customers.
- Security-focused work: authentication and access control with Spring Security, SAML and JWT.
- Tuned complex Oracle and SQL Server queries and IBM Tivoli batch workflows.

Technologies: Java, Spring Boot, Spring Security, SAML, JWT, Oracle, SQL Server, IBM Tivoli

Senior Specialist

SABIS, for Banco Sabadell · Jun 2014 – Jun 2022

- Domain expert for Seneca, the core-banking application that produces mortgage payment simulations and amortization schedules.
- Application Manager for Seneca: defined the work and assigned it to an NTT DATA team.
- Development and maintenance of ATM channel services.
- Java 8 on Proteo, the bank's in-house enterprise framework, alongside Spring MVC and Hibernate.

Technologies: Java 8, Proteo, Spring MVC, Hibernate, Oracle

Analyst

Banco Sabadell · Nov 2012 – Jun 2014

- Internal business applications on the Spring ecosystem.

Technologies: Java, Spring, Thymeleaf, Oracle

Analyst & Developer

Caja Mediterráneo (CAM) · Mar 2007 – Nov 2012

- Online banking applications, ATM services and core banking migration projects.

Technologies: Java EE, Oracle

Web Developer

HDV · Dec 2005 – Mar 2007

- Dynamic web applications.

Technologies: PHP, MySQL, JavaScript

Personal Projects

Forma

Personal Fitness Operating System · forma.backendtothefuture.com

A health and fitness planning system that connects body composition, training, nutrition and shopping cost into one weekly decision loop.

- Hexagonal backend with a framework-free domain: JDBC and Flyway over PostgreSQL, no ORM, decided and recorded in an ADR.
- LLM-generated nutrition plans are fenced by a validated contract: a plan may only reference food identifiers the API publishes, and anything the model invents is rejected on import.
- Spec-driven delivery with 13 ADRs, Vitest and Playwright suites gated in CI.

Technologies: Java 21, Spring Boot 3, PostgreSQL, Flyway, React 19, Playwright

Akademia

RAG-Powered Learning Platform · akademia.backendtothefuture.com

Study platform whose core is a RAG pipeline that turns the material a user uploads into tests and questions grounded in the source. Includes spaced-repetition flashcards and a Stripe store.

Technologies: Java, Spring Boot, PostgreSQL, RAG, OpenRouter, Stripe

TokenMeter

Tokenization & LLM Cost Exploration · tokenmeter.backendtothefuture.com

Measures what LLM code generation really costs, comparing agentic loops with single-shot flows, using the jtokkit o200k_base encoder instead of a character-based estimate.

Technologies: Java, Spring Boot, Docker, Cloudflare

Education

Master's Degree in Web Applications and Services Development

Universidad de Alicante · 2010 – 2011

Certificate in Java Enterprise Development

Universidad de Alicante · 2009 – 2010

Computer Engineering Degree

Universidad de Alicante · 1998 – 2005

Equivalent to MSc level under the Bologna Process.

Community

- Author of Backend to the Future, a blog on software architecture, backend engineering and applied AI.
- Hands-on experimentation with local LLMs (Ollama, LM Studio) and agentic development workflows.
- Former mentor for Computer Science students through Fundación Banco Sabadell.

Learning Focus

Advanced Kubernetes · Observability · SRE Principles · AI Agents

Languages

Spanish: Native · English: Full Professional Proficiency (C1)